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# [123] CSE

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## PART C: 16-MARK ESSAY & SYSTEM DESIGN QUESTIONS

*Comprehensive Step-by-Step Architectural Breakdowns, Diagrams & Evaluated Solutions*

**Total Questions: 2 • Marks per Question: 16 Marks • Total Marks: 32 Marks**

**INDEX OF COVERED TOPICS & LEARNING UNITS:**

1. Link Budget Analysis and Power Optimization • 2. Bent-Pipe Transponders and Satellite Interference

**QUESTION 1** Topic: Link Budget Analysis and Power Optimization **[16 Marks]**

### Q1. Analyze the components of the Link Budget Analysis equation and evaluate the impact of free space transmission loss and high-power amplifier non-linearities in a geostationary satellite link carrying a C-band transponder.

Difficulty: HARD | Bloom: Analyze

**[SOURCE GROUNDING]: sc unit 3 (Page 7) — "A geostationary satellite carries a C-band transponder which transmits 20 watts into an antenna with an on-axi..."**

**COMPREHENSIVE 16-MARK EVALUATED SOLUTION & TECHNICAL BREAKDOWN**

#### 1. Theoretical Introduction & Problem Motivation

Satellite communication systems rely heavily on precise link budget calculations to ensure reliable signal transmission between earth stations and spaceborne transponders. Information from the transmitting earth station is sent to the satellite through the uplink, encountering various losses and non-linear transformations along the path. Understanding the received carrier power equation and the effects of components such as High Power Amplifiers (HPAs) is vital for optimal design.

#### 2. Architectural Principles & Step-by-Step Methodology

**2.1 Link Budget Analysis and Received Carrier Power Equation**

The link budget evaluates all power gains and losses from the transmitter, through the medium, to the receiver. The received carrier power C is expressed mathematically taking into account transmission powers, antenna gains, wavelength, distance, and various system losses.

- Calculates received carrier power using transmitting power, transmitter gain, and receiver gain.
- Incorporates wavelength and slant range to account for geometric spreading.
- Accounts for feeder, atmospheric, and free space propagation losses.

**2.2 Free Space Transmission Loss and HPA Non-linearity**

Free space transmission loss represents the signal power loss as electromagnetic waves propagate through space without a physical medium. Furthermore, HPAs operating near saturation exhibit non-linear distortion, requiring Output Back-Off (OBO) to maintain signal integrity.

- Free space loss depends directly on frequency and distance.
- Non-linearities in HPAs cause intermodulation distortion and spectral regrowth.
- Output Back-Off (OBO) reduces operating output power below saturation power to ensure linearity.

#### 3. Architectural Schematic & Visual Model Representation

**[Diagram] Satellite Uplink and Downlink Power Flow with Transponder Back-Off**

Schematic showing earth station transmission, uplink path, satellite HPA with back-off, and free space propagation.

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[Earth Station] ---> (Uplink: L_s, L_atm) ---> [Satellite HPA (OBO)] ---> [Transponder]
```

#### 4. Mathematical Formulations & Worked Numerical Proofs

**Case 1: Geostationary Satellite C-Band Transponder Power Analysis**

Step 1: Step 1: Identify transmitted power P_T = 20 Watts and on-axis antenna gain G_T = 30 dB (linear gain = 1000).
Step 2: Step 2: Apply the link budget parameter components to evaluate effective isotropic radiated power (EIRP) = P_T * G_T.

Step 3: Step 3: Calculate EIRP in dBW: 10 * log10(20) + 30 dB = 13.01 + 30 = 43.01 dBW.
**Result: EIRP = 43.01 dBW**

### 5. Critical Trade-offs, Advantages & Operational Invariants

[+] Advantage: Ensures reliable communication links over vast geographical areas.
[+] Advantage: Output Back-Off prevents saturation and maintains signal fidelity.
[-] Limitation: Free space and atmospheric losses severely attenuate high-frequency signals.
[-] Limitation: HPA back-off reduces the maximum DC-to-RF conversion efficiency of the amplifier.

### 6. Synthesis, Conclusion & Practical Real-World Applications

A rigorous link budget analysis integrating free space loss models and proper HPA back-off management is essential for designing resilient geostationary satellite communication systems.

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**QUESTION 2** Topic: Bent-Pipe Transponders and Satellite Interference **[16 Marks]**

## Q2. Examine the operational characteristics of bent-pipe transponders, combined C/N ratios using the inverse sum formula, and the mitigation of satellite circuit interference in shared frequency bands.

Difficulty: HARD | Bloom: Analyze

**[SOURCE GROUNDING]: sc unit 3 (Page 7) — "A geostationary satellite carries a C-band transponder which transmits 20 watts into an antenna with an on-axi..."**

### COMPREHENSIVE 16-MARK EVALUATED SOLUTION & TECHNICAL BREAKDOWN

### 1. Theoretical Introduction & Problem Motivation

Modern satellite networks frequently employ bent-pipe transponder architectures to relay signals from uplinks to downlinks with frequency translation but without onboard demodulation. When many satellite circuits and transponders operate within the same frequency band, complex engineering challenges arise regarding noise accumulation and inter-circuit interference.

### 2. Architectural Principles & Step-by-Step Methodology

#### 2.1 Bent-Pipe Transponder Architecture and Combined C/N Ratio

A bent-pipe transponder simply receives, amplifies, and retransmits the RF signal. Because noise is introduced on both the uplink and downlink paths, the overall performance of the link is evaluated by combining individual carrier-to-noise ratios.

- Uplink noise and downlink noise combine to degrade overall link quality.
- The combined C/N ratio of a bent-pipe transponder satellite link is determined using the inverse sum formula.
- Transparently translates frequencies without processing digital baseband data.

#### 2.2 Satellite Circuit Interference in Shared Frequency Bands

When numerous satellite circuits and transponders are deployed within identical frequency bands, signal leakage and spatial overlap create significant cross-circuit interference, necessitating robust frequency planning and polarization techniques.

- Co-channel interference degrades system throughput and bit error rate.
- Interference management requires careful antenna beam shaping and spatial isolation.
- Cross-polarization discrimination (XPD) helps segregate overlapping signals.

### 3. Architectural Schematic & Visual Model Representation

**[Diagram] Bent-Pipe Transponder Link Noise Interaction**
Flow diagram showing uplink C/N_up, transponder translation, downlink C/N_down, and combined output C/N_total.

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[Uplink C/N_up] ---> [Bent-Pipe Transponder] ---> [Downlink C/N_down] ---> [Combined (C/N)_total]
```

### 4. Mathematical Formulations & Worked Numerical Proofs

**Case 1: Inverse Sum Formula Calculation for Bent-Pipe C/N Ratio**

Step 1: Step 1: Define uplink carrier-to-noise ratio (C/N)_up = 20 dB (linear ratio = 100).
Step 2: Step 2: Define downlink carrier-to-noise ratio (C/N)_down = 13 dB (linear ratio = 20).
Step 3: Step 3: Apply the inverse sum formula: 1 / (C/N)_total = 1 / (C/N)_up + 1 / (C/N)_down = 1/100 + 1/20 = 0.01 + 0.05 = 0.06.
Step 4: Step 4: Compute final (C/N)_total = 1 / 0.06 = 16.67 (approx 12.22 dB).
**Result: (C/N)_total = 12.22 dB**

### 5. Critical Trade-offs, Advantages & Operational Invariants

[+] Advantage: Bent-pipe transponders are highly reliable, cost-effective, and format-independent.
[+] Advantage: Enables transparent relay of multiple modulation schemes.
[-] Limitation: Noise accumulates from both uplink and downlink paths.
[-] Limitation: Vulnerability to inter-circuit interference when sharing frequency bands.

### 6. Synthesis, Conclusion & Practical Real-World Applications

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Bent-pipe transponders offer transparent and robust signal relay, but careful calculation of combined C/N ratios via the inverse sum formula and mitigation of co-channel interference are mandatory for optimal network performance.